BRIAN KAGIRI NJUGUNA

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PROFESSIONAL SUMMARY

Senior Systems Engineer, AI Specialist, and Multidisciplinary Technologist with deep expertise in distributed backend architectures, full-stack web applications, and technical AI evaluation workflows. Experienced in building high-throughput, latency-sensitive Python, JavaScript/TypeScript, and Swift infrastructures alongside hands-on graphics rendering, dataset engineering, and simulation. Seamlessly bridges core software engineering with spatial data analysis (GIS), scientific data visualization, parametric CAD design, and interactive media. Highly skilled at leading remote, cross-functional technical teams, designing technical documentation, and deploying robust data visualization pipelines to build scalable, production-grade systems in demanding environments.

CORE SKILLS

Systems & Engineering: Software Architecture, Data Structures, Algorithms, Distributed Infrastructure, High-Throughput Databases, Performance Profiling, Legacy Migration, Batch Processing, CI/CD, Automated Testing

AI & Dataset Engineering: LLM Evaluation, Prompt Optimization, Natural Language Processing (NLP), RLHF Pipeline Integration, Structured Technical Annotation, Synthetic Dataset Generation, Human-in-the-Loop Frameworks

Geospatial & CAD Engineering: GIS Mapping, Terrain Reconstruction, Spatial Database Analysis, Parametric CAD Modeling, Mechanical Assemblies, Electronics Design Automation (EDA), PCB Layout & Simulation

Scientific & Creative Visualization: Medical Image Rendering (DICOM), Scientific Data Visualization, 3D Fluid/Volume Simulation, UI/UX Design, Interactive Game Engine Architectures, Audio-Visual Synchronization

TECHNICAL TOOLS

Programming Languages: Python, JavaScript, TypeScript, Swift, SQL (PostgreSQL, MySQL, SQLite), HTML5, CSS3

Frameworks & Mobile: Next.js, React, Tailwind CSS, Redux, Node.js, FastAPI, Flask, SwiftUI, UIKit, iOS SDK

Cloud, Database & Infra: AWS, GCP, Docker, Git, Redis, MongoDB, GitHub Actions, Unix Systems

Geospatial & Scientific Rendering: QGIS, ParaView, 3D Slicer, GPS Processing, Spatial Analyst, DICOM Segmentation

Engineering, Graphics & Audio: FreeCAD, KiCad, Ngspice, Blender, GIMP, Inkscape, Krita, Lightworks, OpenShot

Game Development: Godot Engine, Defold, Solar2D

PROFESSIONAL EXPERIENCE

SeniorFull-Stack/AIEngineer
HandshakeAI
2025–Present
UnitedStates

• Architectedandmaintaineddistributedbackendservicessupportinghigh-throughputprocessing for automated AI evaluation pipelines.

• Developed asynchronous, low-latency REST APIs using FastAPI, Next.js, and Redis, scaling write-heavy operations.

• Engineered caching layers and decoupled heavy background jobs using Redis-backed queues, yielding substantial latency reductions.

• Designed structured model evaluation frameworks, improving LLM output accuracy, prompt durability, and security alignments.

Full-StackSoftwareEngineer *2024–Present*
OutlierAI *UnitedStates*

- BuiltreactiveinternaldashboardsforvisualizationofLLMtrainingoutputsandinteractivehuman preference data collection.
- Builtscalabledatabaseschemasandautomatedworkflowstotrackgradingtrendsandprogrammatic evaluation annotations.
- OptimizedSQLindexing,querystructures,andasynchronousbackgroundworkerthreadstomanage massive volumes of dataset records.
- Analyzed, profiled, and benchmarked code snippets to isolate and remediate memory leaks and performance regressions.

AITraining&TechnicalProjectSpecialist *2023–Present*
FreelanceRemoteContractor *Remote*

- Engineeredhigh-fidelitydatasetprojectsandspatial,physical,andmechanicalsimulationmodels used for training multimodal AI.
- DesignedGISmappingpipelinesandadvancedterrainanalysisprojectsutilizingQGISandspatial dataset overlays.
- DevelopedcustomParaViewscientificvisualizationprojectsandreconstructed3Dmedicaldatasets from DICOM sources.
- BuiltparametricCADprototypes,mechanicalassemblies,andtechnicalmechanicalschematicsin FreeCAD.
- Authoredclean,exhaustivetechnicaldocumentation,workflowguides,andstep-by-stepverification protocols.

SoftwareEngineer *2023–2024*
ScaleAI *UnitedStates*

- Developed high-throughput data processing and orchestration pipelines to handle massive ML training datasets.
- Implemented parallel processing techniques to speed up heavy file parsing and dataset validation.
- Built internal automation tooling to test, deploy, and monitor backend evaluation workloads.

MachineLearningEngineer *2022–2023*
IBM *UnitedStates*

- Scaled machine learning model serving microservices within Kubernetes, ensuring low response latency.
- Integrated robust CI/CD routines for automated linting, unit testing, and deployment of natural language processing services.
- Cleaned, transformed, and managed complex enterprise-scale databases, providing refined assets for training pipelines.

KEY PROJECTS & HIGHLIGHTS

Full-Stack SaaS Platform

- Builtamulti-tenantplatformfeaturingsecureuserauthentication,optimizedrelationaldatabase queries, and distributed caching.
- Technologies used: Next.js, Node.js, PostgreSQL, Redis, and Tailwind CSS.

AI Evaluation & Optimization Platform

- Designedcustomizedtestingtoolsforbenchmarkinggenerative AImodels,evaluatingformatting accuracy and performance.
- Standardizedmulti-agentvalidationscripts,automatedpromptpipelines,andcentralizedmetrics dashboards.

iOS Productivity Application

- EngineeredaresponsivemobileapplicationutilizingSwiftandSwiftUI,leveragingmodernMVVM architecture.
- Implemented asynchronous networking, local database synchronization, and multi-thread performance optimizations.

Geospatial Analysis & Scientific Visualization

- Built custom GIS templates for mapping terrain data, processing spatial layers, and generating spatial-analyst maps.
- Programmed DICOM slice segmentations and ParaView workflows to render complex medical and physical volume simulations.

Mechanical CAD Design & PCB Layout

- Developed physical product assemblies using parametric design tools, producing detailed engineering assembly drawings.
- Designed multi-layer circuit layouts, routed PCB signals, and conducted analog signal testing in KiCad and Ngspice.

Multimedia Production & Interactive Simulations

- Programmed interactive environment prototypes, pathfinding systems, and user interfaces in Godot Engine.
- Producedandeditededucationalmultimediavideosutilizingadvancedlineareditors,integrating custom motion graphics and audio cues.

EDUCATION

Degree in Computer Science *2023–2026*
University Of Nairobi

- **Research Focus:** Scalable AI systems, distributed systems architecture, and machine learning pipeline optimization.
- **Dissertation:** Design and optimization of large-scale AI-driven systems.

Bachelor of Science in Software Engineering *2020–2023*
University Of Nairobi *Nairobi, Kenya*

- **Specialization:** Machine Learning, Cloud Computing, and Scalable System Design.

DiplomainComputerScience .
KenyaInstituteofSoftwareEngineering *2021–2023*
Nairobi,Kenya

- **Relevant Coursework:** Software Engineering, Database Systems, Computer Graphics, Networking, Data Structures & Algorithms.

CERTIFICATIONS

Stanford University: Machine Learning | Google: Cloud AI Specialist | AWS: Machine Learning Foundations

IBM: Data Analytics Professional | GIS & Spatial Analysis Fundamentals | CAD Design & 3D Modeling

Video Production & Editing | AI-Assisted Workflow Development | Digital Media & Visualization

SOFT SKILLS

Analytical Thinking | Technical Communication | Collaborative Leadership | Rapid Adaptation | High Attention to Detail | Creative Problem Solving | Systematic Troubleshooting

LANGUAGES

English: Professional Proficiency

Swahili: Native Proficiency

ADDITIONAL INFORMATION

- Available for Remote Contract Opportunities.
- Experienced with Cross-Platform Technical Tools and Operating Systems.
- Highly comfortable working in mixed creative, mechanical, and software environments.
- Solid operational understanding of AI-assisted asset production and content pipelines.